

Bedwetting Alarm

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Bedwetting is one of the most common childhood problems that can have a significant impact on a child's health if left unattended. During the night, the parents would not know if the child has urinated and the kid may sleep all night long on a wet bed. Bedwetting alarm provides a very effective solution to this problem. The alarm will sound

as soon as the child begins to wet the bed.

The alarm can also be very helpful for physically challenged people, hospital patients and people suffering from nocturnal enuresis. The bedwetting alarm's special moisture sensor (wet sensor) placed on the bed sheet near the child triggers an alarm at the start of urination. The alarm's sound is loud enough to awaken the child and the parents, who can take the child to the bathroom and the wet clothes of the kid can be changed before going back to sleep.

A thin cable runs from the moisture sensor to the audio alarm unit, which is attached with a safety pin to the bed sheet as shown in Fig. 1.

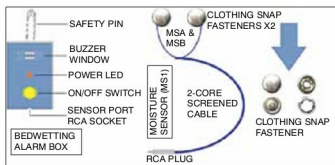


Fig. 1: Alarm and sensor units

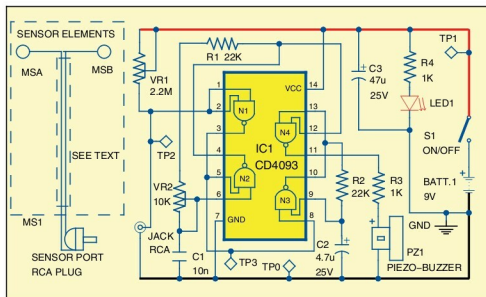


Fig. 2: Circuit of bedwetting alarm



Circuit and working

The circuit is a battery-operated audio alert signal generator built around a popular quad two-input CMOS NAND gate with Schmitt trigger IC CD4093 (IC1). The circuit of the bedwetting alarm is shown in Fig. 2.

In this circuit, gate N1 of IC1 is used to check the status of moisture sensor MS1. Variable resistor VR1 works as a sensitivity-control preset. If the sensor is wet, the input of N1 drops to low level. A gated oscillator is implemented to produce a pulsed tone to drive the piezoelectric audio alarm using gates N2, N3 and N4. Current consumption of this circuit on standby is very low, thus allowing the use of a 9V compact battery to drive the circuit for many months.

A pair of commonly available clothing snap fasteners (MSA and MSB) can be used as moisture sensor MS1. Attach the sensor elements to the bed sheet such that the bed sheet is between MSA and MSB.

Working of the circuit is simple. When the child urinates, the bed sheet becomes wet allowing conduction be-

Test Points

Test point	Details
TP0	0V, GND
TP1	9V
TP2	Low (when the bed is wet)
TP3	High (when the bed is wet)

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